

## **Final Paper**

**Game Market Pulse: Visualizing Video Game Sales, Genre Trends, and Critical  
Reception (1980 to 2024)**

**DASC 5231**

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## 1. Introduction and Project Objective

The video game industry has grown from a small hobby into one of the largest entertainment industries in the world. Over the past 40 years, it has undergone major transformations. It began with arcade machines and early home consoles, developed into a strong era of physical game sales in the 2000s, and is now largely driven by digital downloads and mobile gaming. Alongside these shifts, the ways games are developed, distributed, and consumed have also changed significantly.

This project analyzes video game sales data from 1980 to 2024 to better understand what drives a game's success. The main goal is to answer key questions about how the industry has evolved and what factors influence commercial performance. The first objective is to analyze the structural shift in the video game market over time, with a focus on how the relationship between game releases and sales has changed alongside the rise of digital and mobile platforms. The second objective is to identify the main drivers of commercial success by comparing performance across platforms, genres, and publishers, and highlighting patterns of market dominance. The third objective is to examine the relationship between game quality and sales by analyzing critic scores and evaluating whether higher-rated games consistently achieve stronger commercial outcomes. The fourth objective is to assess market efficiency by comparing average sales per release across genres and decades in order to identify which categories generate the highest returns and how this changes over time.

Overall, this study aims to provide a clear and structured understanding of the video game industry by examining it from multiple analytical perspectives. Breaking the data down by time period, platform, and game characteristics, it highlights the key factors that contribute to commercial success and how the industry has evolved over time.

## 2. Dataset description

The dataset used in this project is called *Video Game Sales and Industry Data (1980–2024)*, published by Bhushan Divekar on Kaggle. It includes approximately 18,000 video games released between 1980 and 2024 and is used to study trends in the video game industry over time. Fortunately, the dataset had already been cleaned before loading it into Tableau. The dataset contained 18,832 rows across eight columns. This included basic information about each game, such as title, console, genre, publisher, developer, and release year. It also includes critic scores, which indicate how well a game was reviewed and can serve as a simple measure of quality. It should be noted that the cleaned dataset did not include the regional sales breakdown, which was in the raw dataset, nor did it include digital sales or mobile applications. These gaps are further discussed in the limitations section of this report.

This dataset is useful because it covers a long time period and many parts of the gaming market. However, it also has some limits. It mostly includes physical game sales and does not fully include digital sales, which became very important after 2010. The dataset also does not include mobile games, which are a big part of the industry today. Because of this, the most recent years in the dataset may not show the full size of the modern gaming market.

### 3. Methods and tools used

The goal of this analysis is to explore patterns in video game sales in a way that is clear and easy to understand for any viewer, whether a stakeholder or a consumer. This project focuses on building visualizations in Tableau and combining them into an interactive dashboard with filters. We chose Tableau for this analysis because it can handle large datasets and allows users to interact with the data through filters and multiple chart views.

All visualizations were created in Tableau using the cleaned dataset and then combined into a single dashboard that followed a concise narrative structure. The first part of the structure was the market context, followed by category-level performance across consoles, genres, and publishers, then an examination of the relationship between critic scores and sales, and ending with a market efficacy analysis. Global filters for release year and genre were also added and linked across all sheets so that if one selection is made in one view, it would automatically update the others. This makes it easier for viewers to compare trends and focus on specific parts of the dataset.

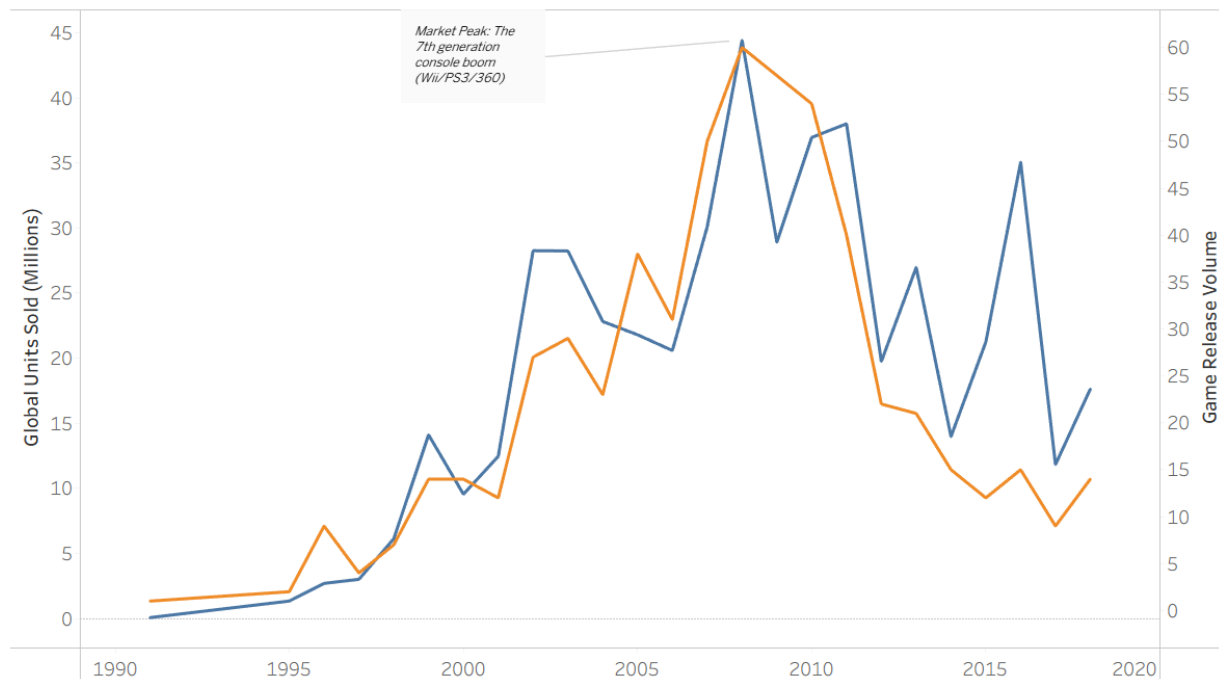
### 4. Visualization Design and Rationale

The dashboard follows a logical progression: first establishing the historical market context, then examining performance across genres, and finally applying statistical modeling to evaluate the relationship between product quality and commercial success.

#### Evolution of the Global Video Game Market (1980–2020):

This chart uses a dual-axis time series to compare **demand** (global units sold in millions) and **supply** (number of game releases) from 1980 to 2024. Raw data is used instead of a smoothed average to preserve the volatility typical of a hit-driven entertainment market.

Evolution of the Global Video Game Market (1980–2020)



### Key insights:

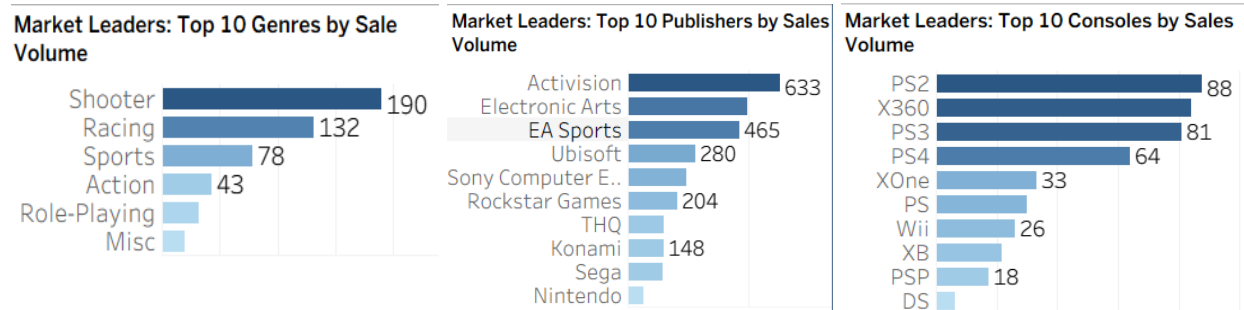
- **The “Golden Age” (2005–2010):** The era of Wii, PS3, and Xbox 360 marks the peak of the physical retail market. Supply and demand rise together, indicating a highly synchronized period of growth.
- **Shift in correlation:** Between 1990 and 2008, a strong positive relationship exists between the number of games released and total sales. After 2010, this relationship got weaker, suggesting structural changes in the industry.
- **The 2016 paradox:** A secondary spike in release volume appears around 2016, while global sales continue to decline. This indicates that production remained high even as the importance of physical retail decreased.

### Data limitations:

- **Digital sales gap:** The dataset primarily reflects physical sales. The decline after 2010 likely aligns with the shift toward digital platforms, which are not fully captured.
- **Mobile gaming impact:** The rise of smartphones from around 2008 represents a major shift in consumer behavior, but this trend is not visible in a console-focused dataset.

### Top Performance by Category:

Three bar charts (Consoles, Genres, and Publishers) present the top 10 performers in each category. Applying a Top-N filter reduces noise from smaller contributors and highlights the main sources of market share.



### Key insights:

- **Console dominance:** PS2 and Nintendo DS lead in total sales, reflecting the impact of a large install base on software performance. This aligns with the growth observed between 2000 and 2010.
- **Genre leaders:** Action and Sports genres significantly outperform others, indicating broad and consistent consumer demand.
- **Publisher concentration:** A small number of publishers (Nintendo, EA, and Activision) account for a large share of total sales, highlighting the importance of scale, marketing, and distribution capabilities.

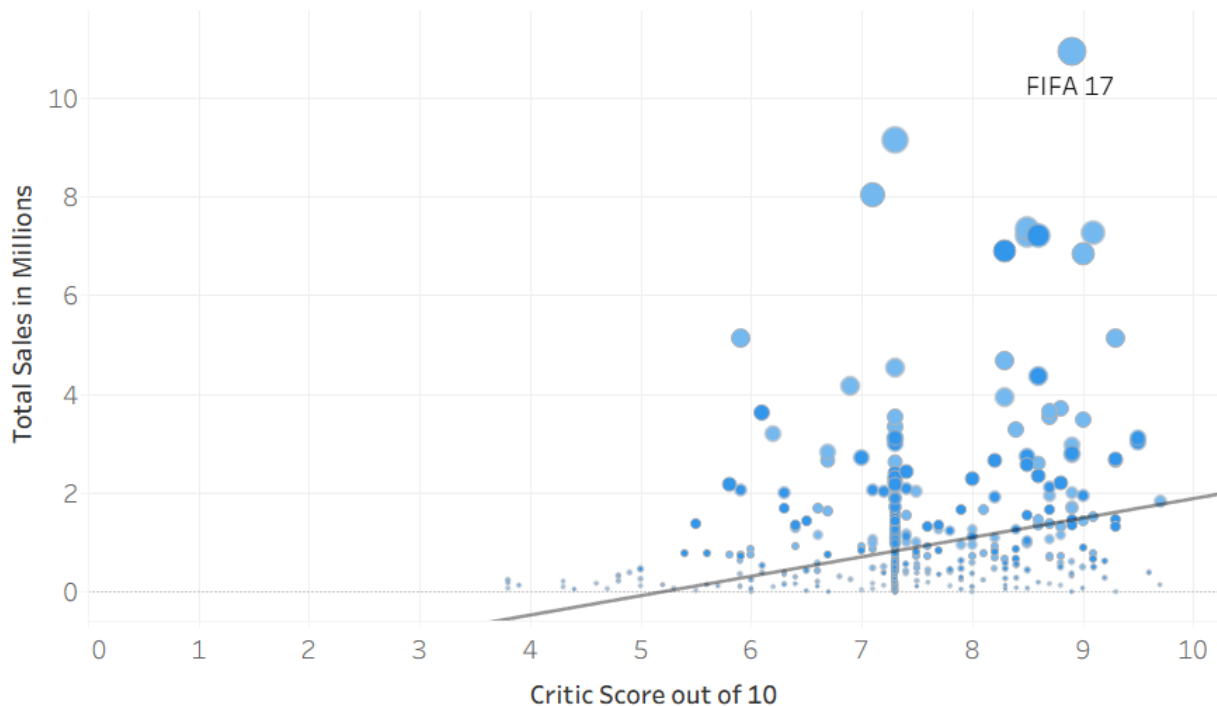
### Data limitations:

- **Legacy bias:** Cumulative totals from 1980 to 2024 favor older platforms and genres, making newer systems like the Nintendo Switch appear comparatively weaker.
- **Publisher changes:** Mergers and acquisitions are not reflected, meaning actual industry concentration is likely higher than shown.

### Critic Score vs. Sales:

This scatter plot, combined with a linear regression model, examines the relationship between critic scores and sales. Data points are grouped into “Market Leaders” (Action, Sports, Shooters) and other genres to assess the influence of category popularity.

**Critic Score vs Sales:** Higher Scores Correlate with Increased Sales ( $P < 0.0001$ )



#### Key insights:

- **Quality and sales link:** The regression model shows that a 1-point increase in critic score is associated with approximately 340,000 additional units sold, indicating a measurable relationship between quality and commercial performance.
- **Genre advantage:** Titles in high-demand genres tend to appear above the regression line, suggesting that category popularity amplifies sales outcomes.
- **7.5 score threshold:** Sales increase more sharply once critic scores exceed approximately 7.5, indicating a practical benchmark for strong commercial performance.

#### Data limitations:

- **Survivorship bias:** The dataset is skewed toward games that were widely distributed and reviewed, excluding many low-performing or niche titles.
- **External influences:** Factors such as social media trends, influencer marketing, and viral popularity are not captured in critic scores but can significantly affect sales.

#### Market Efficiency: Average Sales per Release by Decade:

A highlight table presents average sales per game to evaluate efficiency across genres and time periods. This approach shifts the focus from total volume to per-title performance, helping identify which genres deliver the highest return per release.

### Market Efficiency: Average Sales (\$ Million) per Release by Decade

|              | 1970s | 1980s | 1990s | 2000s | 2010s | 2020s |
|--------------|-------|-------|-------|-------|-------|-------|
| Action       | 0.55  | 0.47  | 0.57  | 0.40  | 0.37  | 0.07  |
| Racing       | 0.27  | 0.73  | 0.50  | 0.38  | 0.29  | 0.04  |
| Sports       | 0.15  | 0.58  | 0.46  | 0.40  | 0.57  |       |
| Misc         |       | 0.26  | 0.38  | 0.31  | 0.22  | 0.07  |
| Shooter      | 0.98  | 0.95  | 0.34  | 0.45  | 1.11  |       |
| Role-Playing |       | 1.09  | 0.28  | 0.25  | 0.32  | 0.32  |

#### Key insights:

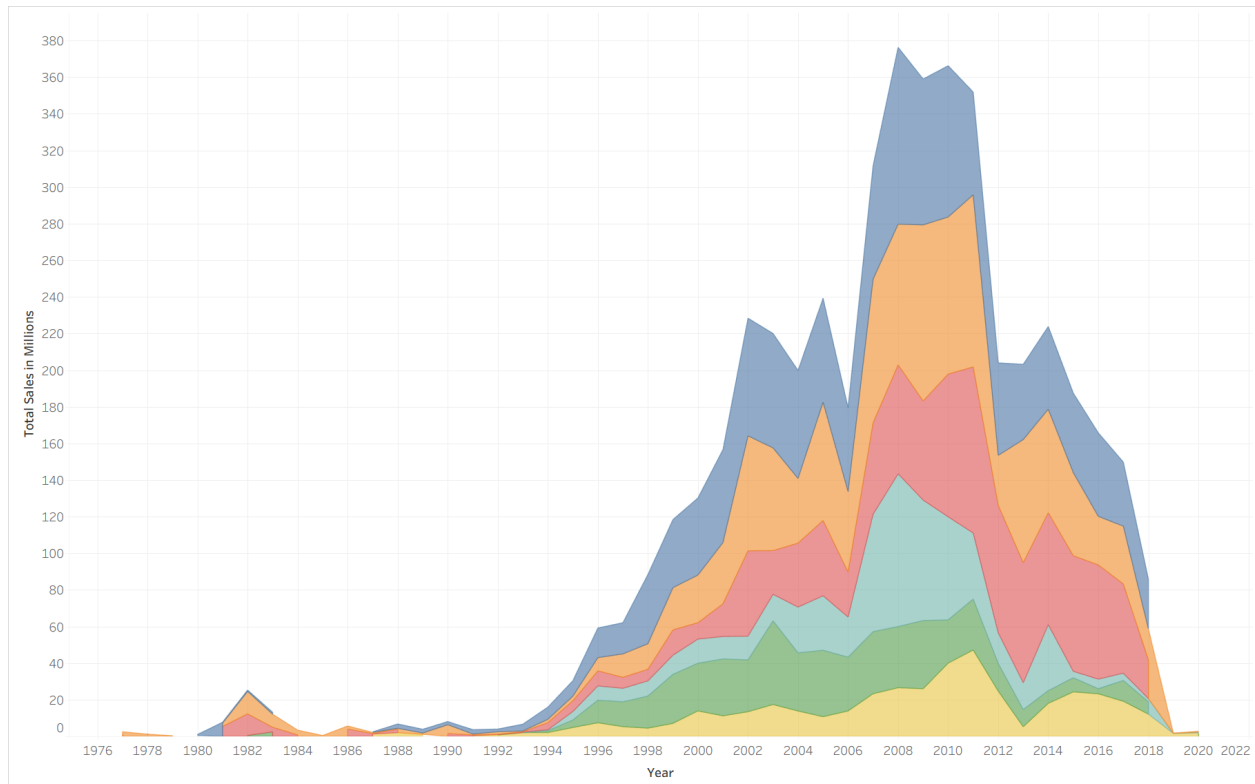
- **Blockbuster trend (2010–2020):** Genres such as Shooters and Sandbox games show high average sales, reflecting a shift toward fewer, high-impact releases.
- **Early genre strength:** Platform and Role-Playing games demonstrate high efficiency in earlier decades, when competition was lower and these genres defined the market.
- **Niche limitations:** Strategy, Adventure, and Puzzle genres consistently show lower average sales, indicating smaller but stable audience segments.

#### Data limitations:

- **Outlier effects:** A single highly successful title can significantly influence average values for a genre or decade.
- **Cost considerations:** Revenue-based efficiency does not account for development costs; high-selling games may still have lower profit margins due to large production budgets.

#### Market Share Over Time

This is a stacked area chart that examines how various game genres contributed to the total sales from 1975 to 2020. A filter was added to display the top 6 genres by total sales: Sports, Action, Shooter, Miscellaneous, Racing, and Role Playing. Displaying every genre would cause clutter.



### Key Insights:

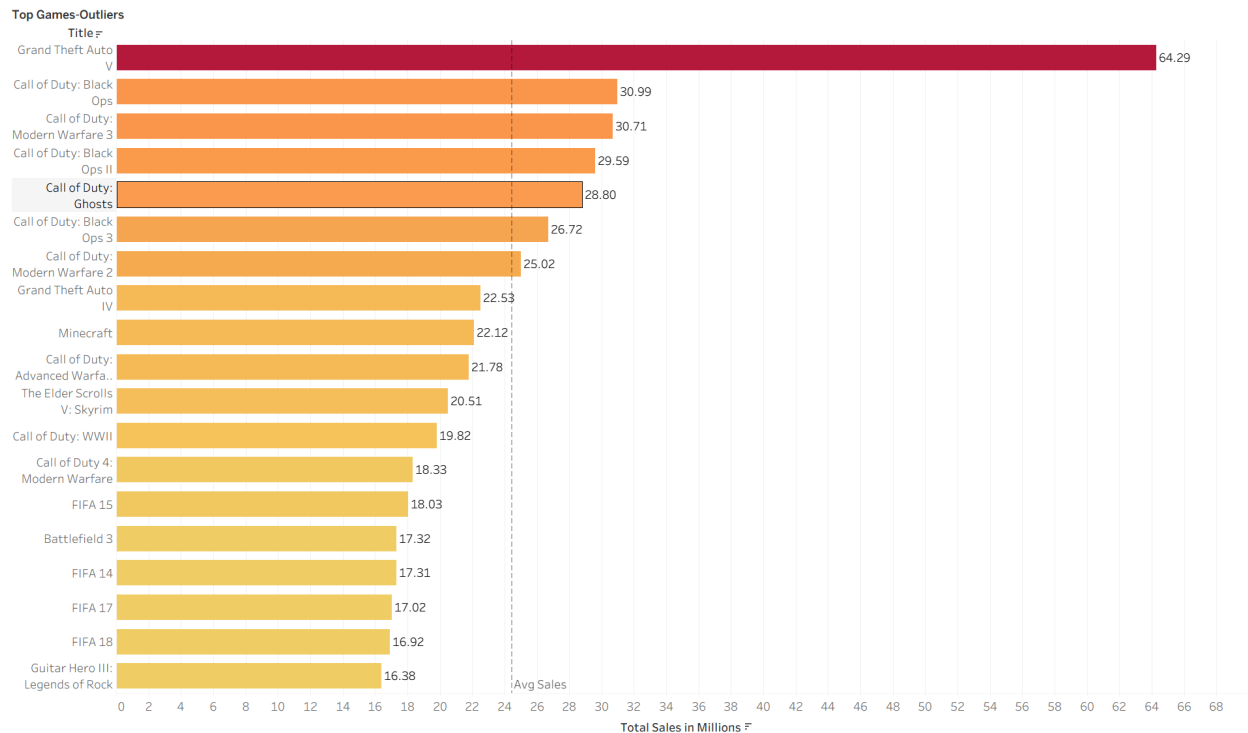
- **Genre Dominance:** The largest bands throughout the chart were undoubtedly Sports and Action, signifying that these two genres were commercially dominant in the entire history of gaming
- **Industry Peak:** There was a sharp peak in total sales across all genres between 2008 and 2011; this was also when consoles like PS3, Wii, and Xbox 360 were at their peak in popularity.
- **Surge in Shooter:** There was a noticeable surge in shooter games from the mid-2000s onwards. This is due to franchise games like Call of Duty and the growing popularity of online gaming platforms.
- **Post-2012 Decline:** There is a noticeable, steep drop in sales after 2012, which could signal an industry shift towards digital game distribution.

### Data Limitations:

- **Digital Sale Gap:** Because this dataset only captures sales of physical games, there is a gap for digital sales, which were the most popular method of gaming distribution during this time.
- **Genre Classification:** Band sizes may vary, as some games may belong to more than one genre, depending on how the publisher-assigned genre labels are applied in the dataset.

## Top Games - Outliers

This is a horizontal bar chart that shows the best-selling games by total sales, with a filter applied to show only the top 20 titles. A color gradient was added to indicate sales magnitude, along with a reference line marking the average sales threshold. This visualization was designed to identify true statistical outliers.



## Key Insights:

- **GTA V:** Without a doubt, GTA V is the clear statistical outlier with 64.29 million sales and stands significantly apart from other top performers in this chart.
- **Franchise Dominance:** The Call of Duty franchise appears in seven of the 20 entries, demonstrating that if a franchise is consistent and releases games annually, it will clearly show in the sales figures.
- **Average Sales Threshold:** The reference line falls around 24-25 million, which means that only the top 7 games meaningfully cross it.
- **Top Genres:** This visualization confirms that the top genres are action/shooter.

## Data Limitation

- **Recency Bias Gap:** Titles published after 2018 are underrepresented, possibly because their sales data are incomplete in the dataset.

## KPI Summary

| Total Sales | # of Games | Top Platform |
|-------------|------------|--------------|
| 4,814.13M   | 11,779     | PS2          |

The KPI summary provides quick snapshots of the video game dataset, which provides viewers with context before looking further at the dashboard. Three important metrics were displayed: Total Sales (in millions), Total Number of Games, and Top Platform by Sales.

### Key Insights:

- **Market Scale:** The total sales figure represents the global sales of games captured in the dataset. It reflects how large the gaming industry has grown over the tracked period of time (1970-2020).
- **Top Platform:** PlayStation 2, the top platform by total sales, has been consistent historically.
- **Dataset Size:** The total number of games in the dataset indicates its diversity, spanning a wide range of genres, decades, and consoles. This gives statistical credibility to the other visualizations.

## 5. Limitations

One major limitation is that the dataset mostly includes physical game sales only. Around 2010, the industry started shifting toward digital platforms like Steam and console online stores. By the mid 2010s, digital sales became the main way people buy games. Because this data does not include digital sales, the drop in sales after 2012 likely reflects the decline of physical copies rather than a real decline in the overall gaming industry.

Another gap is that mobile gaming is not included. Since smartphones became popular, mobile games have become one of the largest parts of the industry. Games like Candy Crush or Pokémon GO reach massive audiences, but they are not captured in this dataset. This means the analysis mainly reflects console and PC games and does not fully represent the modern gaming market, especially in recent years.

Finally, there is also a bias toward older platforms and publishers. Since the dataset covers more than 40 years, older consoles like the PlayStation 2 naturally appear higher in total sales because they have had more time to accumulate results. This can make newer platforms look weaker even if they are performing strongly in today's market.

## **6. Future Improvements**

One important improvement would be to include digital sales and mobile gaming data. Because the current dataset mainly reflects physical game sales, which means it does not fully capture how the industry works today. Adding data from digital platforms like Steam and console online stores, as well as mobile games, would give a more complete and accurate view of the modern gaming market. This would also help better explain trends after 2010, when digital and mobile gaming became dominant.

Another improvement would be to reduce bias and improve the accuracy of comparisons. This could be done by using more balanced metrics like median sales instead of averages to reduce the impact of extreme outliers. It would also help to normalize results by time or platform lifespan, so older consoles do not have an unfair advantage. These changes would make the analysis more fair and improve how well it reflects real industry performance.

## **7. Conclusion**

The analysis highlights how the video game industry has evolved from a tightly coupled, physical retail market into a more complex and fragmented digital ecosystem. Historically, growth was driven by a strong alignment between supply and demand, particularly during the mid-2000s console boom. However, this relationship weakened after 2010, reflecting major structural shifts such as the rise of digital distribution and mobile gaming.

Across categories, success is not evenly distributed. A small number of consoles, genres, and publishers dominate overall market share, showing that scale and established ecosystems play a critical role in driving performance. At the same time, genre analysis reveals that broad-appeal categories like Action and Sports consistently outperform others, while niche genres tend to operate within more limited commercial ceilings.

The statistical analysis further confirms that product quality has a measurable impact on sales, with higher critic scores generally leading to better commercial outcomes. However, quality alone is not sufficient—genre selection and market demand significantly influence results. Games in high-demand categories benefit from an additional “genre advantage,” often outperforming similarly rated titles in less popular segments.

Finally, the shift toward higher average sales in certain genres during recent decades suggests a move toward a “blockbuster” model, where fewer but larger releases generate the majority of revenue. This indicates increasing market consolidation and higher barriers to entry, particularly for large-scale productions.

Overall, the findings suggest that success in the modern video game industry depends on a combination of factors: aligning with high-demand genres, maintaining strong product quality,

and leveraging scale in distribution and marketing. At the same time, the limitations of the dataset—particularly the lack of digital and mobile data—indicate that the current analysis likely underrepresents some of the most significant recent industry trends.